ModelIt!

Student Activity Pack

Potential to Kinetic: The Energy Transformation Chain

How Height, Mass, and Friction Determine Where Energy Goes

Grade 08L2 | NGSS MS-PS3-1, MS-PS3-5 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is potential energy and what is kinetic energy?

2 Where does a roller coaster have the most energy — at the top or bottom of a hill?

3 What happens to a ball's speed when it rolls down a ramp? Where does the speed come from?

4 Draw an energy diagram for a ball rolling down a ramp, showing PE and KE at the top, middle, and bottom.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Height | Mass | Potential Energy | Kinetic Energy | Thermal Energy (Friction)

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Ideal Drop

High height, moderate mass, no friction

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heavy vs. Light

Same height, different masses

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: High Friction

Same starting conditions plus heavy friction

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Air Resistance — Drag force that opposes motion and converts kinetic energy to thermal energy

Elastic Energy — Energy stored in compressed or stretched materials like springs

Sound Energy — Energy converted to vibrations in the air when objects collide or scrape surfaces

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Where does the energy go when a roller coaster slows down at the bottom of a hill? Use evidence from your simulation to support your answers.

1. **Why can't a roller coaster's second hill be taller than its first?**

2. **If energy is conserved, why do machines always seem to 'waste' energy?**

3. **How does the model prove that friction doesn't destroy energy but transforms it?**

4. **Why does a heavier object rolling down a ramp have more kinetic energy but not necessarily more speed?**

5. **Where does the thermal energy from friction go after it's generated? Does it disappear?**

8 STEM Challenge: Design an Energy-Efficient Ramp System

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YOUR ENGINEERING MISSION

Design a ramp system that maximizes the kinetic energy delivered to a target at the bottom, minimizing energy loss to friction, using evidence from your model.

THE SCENARIO

An engineering firm needs to design a delivery system that uses gravity to move packages from a high warehouse shelf to a packing station below. Your team must design the most energy-efficient ramp, maximizing speed at the bottom.

Engineering Design Thinking

1. WHAT RAMP MATERIAL MINIMIZES FRICTION WHILE MAINTAINING SAFETY?
2. HOW STEEP SHOULD THE RAMP BE FOR MAXIMUM EFFICIENCY?
3. WHAT MASS RANGE CAN YOUR SYSTEM HANDLE EFFECTIVELY?
4. HOW WILL YOU MEASURE THE KINETIC ENERGY AT THE BOTTOM TO PROVE EFFICIENCY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Mechanical Engineers and Energy Systems Designers create machines and systems that efficiently convert and transfer energy. They earn $70,000-$120,000/year and work across industries from

transportation to manufacturing.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 Your model shows that a 5 kg ball released from 10 meters has 490 J of potential energy at the top and 490 J of kinetic energy at the bottom on a frictionless ramp. On a ramp with friction, the ball reaches the bottom with only 380 J of kinetic energy. Where are the other 110 J?

- **A.** The 110 J were destroyed by friction and no longer exist
- **B.** The 110 J were converted to thermal energy (heat) in the ramp surface and ball through friction
- **C.** The 110 J remain as potential energy because the ball did not fall the full distance
- **D.** The 110 J were absorbed by the air and converted to sound energy only

2 In your model, a 2 kg object and a 8 kg object are released from the same height. The model shows the 8 kg object has four times more kinetic energy at the bottom. A student concludes the heavier object must fall four times faster. Is this correct?

- **A.** Yes, because more kinetic energy always means more speed
- **B.** No. The heavier object has more kinetic energy because KE depends on mass AND speed. Both objects fall at the same speed (without air resistance), but the heavier one has more energy due to its greater mass.
- **C.** Yes, because gravity pulls heavier objects down faster
- **D.** No, because the lighter object actually falls faster due to less air resistance

3 The model shows that at the midpoint of a frictionless ramp, an object has both potential and kinetic energy. Which statement is true about the total energy at this point?

- **A.** Total energy is less than at the top because some energy has been used up
- **B.** Total energy equals the PE at the top, split between PE (remaining height) and KE (gained speed)
- **C.** Total energy is greater than at the top because the object has both PE and KE
- **D.** Total energy cannot be calculated at the midpoint because the object is in motion

4 Based on your model, why can a roller coaster's second hill never be taller than its first hill?

- **A.** Because the train gains weight as it moves, making it harder to climb
- **B.** Because friction and air resistance convert some kinetic energy to thermal energy along the way, leaving insufficient energy to reach the original height
- **C.** Because gravity weakens as the coaster moves farther from the starting point
- **D.** Because the coaster's engine turns off after the first hill

5 **A student claims that a ball rolling on carpet 'loses energy' compared to rolling on a smooth floor. Using model evidence, how should this claim be corrected?**

- **A.** The student is correct because carpet absorbs and destroys kinetic energy
- **B.** The ball does not lose energy. Carpet has higher friction, which converts more kinetic energy to thermal energy. The total energy (KE + thermal) remains the same as on the smooth floor.
- **C.** The student is wrong because carpet actually adds energy through static electricity
- **D.** The ball loses energy on both surfaces equally because gravity is the same